UNIT 1 · TOPIC 2(B)

Animal-Drawn Equipment

Syllabus topic: Functional requirement, principle of working and constructional features of animal and power operated equipment

> Exam tip: animal-drawn equipment is still asked in ARS/NET.

Ploughs and implements

- **Indigenous (desi) plough:** wooden body with an iron share point. It opens a V-shaped furrow without inverting the soil. Draft ≈ **40–60 kgf** for a pair of bullocks.
- **Improved animal-drawn implements:** MB plough (Victory plough), bund former, blade harrow (bakhar), seed-cum-fertilizer drills (2–3 row), bullock-drawn planter, Olpad thresher (disc-harrow-like, for wheat).

Draught power of bullocks

- A pair of good bullocks develops about **0.75–1.0 hp** in total (each **~0.5 hp** sustained).
- A bullock can exert a draft of about **1/10th of its body weight**.

Yokes

- The conventional yoke is the neck/shoulder yoke.
- Improved yokes (collar, three-pad) raise draught efficiency and animal welfare.

For draft, pull and drawbar power, see Topic 2(a).